Higher derivatives

1

a $\frac{dy}{dx} = -10x^{-6}$

b $\frac{d^2y}{dx^2} = 60x^{-7}$

c $\frac{d^3y}{dx^3} = -420x^{-8}$

2

a $\frac{dy}{dx} = 40x^7$

b $\frac{d^2y}{dx^2} = 280x^6$

c $\frac{d^3y}{dx^3} = 1680x^5$

d $\frac{d^4y}{dx^4} = 8400x^4$

e $\frac{d^5y}{dx^5} = 33\,600x^3$

3

a $\frac{dy}{dx} = -8x^3 + 12x^2 - 14x + 8$

b $\frac{dy}{dx} = -142$

c $\frac{d^2y}{dx^2} = -24x^2 + 24x - 14$

d $\frac{d^2y}{dx^2} = -302$

e $\frac{d^3y}{dx^3} = -48x + 24$

f $\frac{d^3y}{dx^3} = -264$

4

a $y'' = 12x^2 - 36x + 16$

b $f''(x) = -\frac{6}{49}x^{-\frac{13}{7}}$

c $f''(x) = 108(3x+2)^2$

d $f''(x) = 24(x+8)^{-4}$

e $y'' = -4\left(7-6x^2\right)^{-\frac{5}{3}}\left(2x^2+7\right)$

f $y'' = -(6-x)^{-\frac{3}{2}}$

g $f''(x) = 20x^3 + 6$

h $f''(x) = -84x^5 + 36x^2 - 24x$

Concavity

5

a B, C, D

b A

6

a $f'(a) < 0, f''(a) > 0$

b $f'(a) > 0, f''(a) < 0$

c $f'(a) > 0, f''(a) > 0$

d $f'(a) < 0, f''(a) < 0$

7

a

i Positive

ii Minimum

iii $f'(x) = 2x - 4$

iv $f''(x) = 2$

v Concave up

b

i Negative

ii Maximum

iii $f'(x) = -2x + 4$

iv $f''(x) = -2$

v Concave down

8

a $f'(x) = 8x + 3$

b $f''(x) = 8$

c

x	0	1	2	3	4
$f'(x)$	3	11	19	27	35

d Gradient is increasing at a constant rate.

e No, because $f(x)$ is a quadratic function, and it is always concave up.

9

a $f'(x) = -6x + 12$

b $f''(x) = -6$

c Gradient is decreasing at a constant rate.

d No, because $f(x)$ is a quadratic function, and it is always concave down.

10

a

i No such x-values

ii $x > -1$

iii The graph is always increasing for $x \geq -1$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

b

i $x > 2$

ii No such x-values

iii The graph is always decreasing for $x \geq 2$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

c

i $x > 0$

ii $x < 0$

iii The graph is always decreasing for $x < 0$ and for $x > 0$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

d

i $x < 0$ or $x > 0$

ii No such x-values

iii The graph is always increasing for $x < 0$ and always decreasing for $x > 0$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

11

a Horizontal translation of 6 units left

b $(0, 0)$

c $(-6, 0)$

d

x	-7	-6	-5
y'	3	0	3
y''	-6	0	6

e Yes, because $f'(-6) = 0$.

f $x > -6$

12

a Horizontal translation of 5 units left

b $(0, 0)$

c $(-5, 0)$

d

x	-6	-5	-4
y'	3	0	3
y''	-6	0	6

e Yes, because $f'(-6) = 0$.

f $x > -5$

13

a $y'' = 12x^2 - 48x$

b $(0, -9), (4, -265)$

c

x	-2	0	2	4	6
y'	-128	0	-96	-128	0
y''	144	0	-48	0	144

d $(0, -9)$ is a horizontal point of inflection and $(4, -265)$ is an ordinary point of inflection.

e $x < 0$ or $x > 4$

14

a $y'' = 24x - 32$

b $\left(\frac{4}{3}, -\frac{206}{27}\right)$

c An ordinary point of inflection

d $x < \frac{4}{3}$

15

a $y'' = 18x + 12$

b $\left(-\frac{2}{3}, -\frac{5}{9}\right)$

c Ordinary

d $x < -\frac{2}{3}$

16

a $y'' = 20x^3 - 6$

b $\left(\sqrt[3]{\frac{3}{10}}, \left(\sqrt[3]{\frac{3}{10}}\right)^5 - 3\left(\sqrt[3]{\frac{3}{10}}\right)^2\right)$

c Ordinary

d $x > \sqrt[3]{\frac{3}{10}}$

e $x < \sqrt[3]{\frac{3}{10}}$

17

a $x < -3$ or $x > 0$

b $-3 < x < 0$

c $f''(x) = 6x + 9$

d $x > -\frac{3}{2}$

e $x < -\frac{3}{2}$

f $x = -3$

g $x = -\frac{3}{2}$

Stationary points

18

a $x = 5$

b $f''(5) = 2$

19

a $(-4,0),(2,-108)$

b $f''(x)=6x+6$

c $(-4,0)$ is a maximum turning point and $(2,-108)$ is a minimum turning point.

d $x<-1$

20

a $y'=4x^3+12x^2$

b $y''=12x^2+24x$

c $(0,2),(-3,-25)$

d $(0,2)$ is a point of inflection, and $(-3,-25)$ is a minimum point.

e $(0,2),(-2,-14)$

f $(0,2)$ is a horizontal point of inflection, while $(-2,-14)$ is an ordinary point of inflection.

g $x<-2$ or $x>0$

21

a $f'(x)=(x-5)(3x-21)$

b $(5,0),(7,-4)$

c $f''(x)=6x-36$

d Local minimum at $(7,-4)$, local maximum at $(5,0)$

e $x<6$

22

a

i $(3,-61),(-3,47)$

ii $(-3,47)$ is a maximum turning point and $(3,-61)$ is a minimum turning point.

b

i $(2,-18),(-2,14)$

ii $(-2,14)$ is a maximum turning point and $(2,-18)$ is a minimum turning point.

c

i $(-4,112),(1,-13)$

ii $(-4,112)$ is a maximum turning point and $(1,-13)$ is a minimum turning point

d

i $(-1,2),(0,3),(1,2)$

ii $(0,3)$ is a maximum turning point, $(-1,2)$ and $(1,2)$ are minimum turning points

23

a

i $x=-\frac{3}{7},-\frac{1}{3}$

ii Maximum turning point at $x=-\frac{3}{7}$ and minimum turning point at $x=-\frac{1}{3}$

b

i $x=1,7$

ii Maximum turning point at $x=1$ and minimum turning point at $x=7$

c

i $x=-2$

ii Maximum turning point.

d

i $x=5$

ii Horizontal point of inflection

e

i $x=5$

ii Horizontal point of inflection.

24

a $x=-\frac{3}{5},-\frac{1}{3}$

b $x=-\frac{7}{15}$

c 2

25

a $y''=30\left(x^2-5\right)\left(x^2-1\right)$

b $\left(-\sqrt{5},0\right),(-1,-64),(1,-64),\left(\sqrt{5},0\right)$

c $\left(-\sqrt{5},0\right),\left(\sqrt{5},0\right)$

26

a

i $(5,4)$

ii $(5,4)$

iii Turning point. The graph changes from decreasing to increasing on either side of $x=5$, but the concavity does not change at $x=5$.

b

i $(4,5)$

ii $(4,5)$

iii The graph changes from decreasing to increasing at $x=4$, but the concavity does not change at $x=4$.

c

i $(8,6)$

ii $(8,6)$

iii The graph changes from decreasing to increasing around $x=8$, but the concavity does not change around $x=8$.

27

a $a = b$

b $a = b$

c They can be any real numbers.

28

a $3a + 2b + 12 = 0$

b $b = -3a$

c $a = 4$

d $b = -12$

29

a $a = -2$

b $c = -48$